SVKM'S NMIMS — MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Re-Examination | Dynamic Systems Modeling | Set 1

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1

Answer the following:

(a) Write the expressions for velocity in polar coordinates (v_r and v_θ components) and state when polar coordinates are preferred over Cartesian. **[0.5]**

Ans:

In polar coordinates:

$v_r = \dot{r}$ (radial component)

$v_\theta = r\dot{\theta}$ (transverse component)

$\rightarrow \vec{v} = \dot{r}\,\hat{e}_r + r\dot{\theta}\,\hat{e}_\theta$

Polar coordinates are preferred when the motion has radial symmetry or when dealing with central-force problems (e.g. planetary motion, pendulums), because the constraint is naturally expressed in r and θ.

(b) Define the Lagrangian L of a mechanical system and state the Euler-Lagrange equation in terms of a generalized coordinate q. **[0.5]**

Ans:

The Lagrangian is defined as $L = T - V$, where T = kinetic energy and V = potential energy.

The Euler-Lagrange equation is:

$\frac{d}{dt}(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$

This yields the equation(s) of motion for the system in terms of the generalized coordinate q.

(c) State the transport theorem for a vector quantity as observed from a fixed (inertial) frame and a rotating frame. Identify the correction term. **[0.5]**

Ans:

The transport theorem states:

$(d\vec{A}/dt)_{fixed} = (d\vec{A}/dt)_{rotating} + \vec{\Omega} \times \vec{A}$

where $\vec{\Omega}$ is the angular velocity of the rotating frame.

The correction term $\vec{\Omega} \times \vec{A}$ accounts for the rotation of the basis vectors themselves as seen from the fixed frame.

(d) Define the damping ratio ζ and state the three response regimes (underdamped, critically damped, overdamped) with their conditions. **[0.5]**

Ans:

The damping ratio $\zeta = c / c_{cr} = c / (2\sqrt{km})$, where c = damping coefficient, k = stiffness, m = mass.

Three regimes:

$\zeta < 1$ → Underdamped: oscillatory with exponentially decaying amplitude.

$\zeta = 1$ → Critically damped: fastest return to equilibrium without overshoot.

$\zeta > 1$ → Overdamped: slow, non-oscillatory return to equilibrium.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

(a) A particle moves in a circle of radius r = 2 m at constant angular speed $\dot{\theta} = 3$ rad/s. Calculate the radial (centripetal) acceleration a_r and the transverse acceleration a_θ. **[0.5]**

Ans:

For circular motion at constant angular speed ($\dot{r} = 0$, $\ddot{r} = 0$, $\ddot{\theta} = 0$):

$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 2 \times 3^2 = -18$ m/s² (directed radially inward)

$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 0 = 0$ m/s²

The magnitude of centripetal acceleration is 18 m/s².

(b) Write the kinetic energy T and potential energy V for a simple pendulum of mass m and length L, using θ as the generalized coordinate. **[0.5]**

Ans:

For a simple pendulum with angle θ:

$T = \frac{1}{2} m (L\dot{\theta})^2 = \frac{1}{2} m L^2 \dot{\theta}^2$

$V = -mgL \cos\theta$ (taking the pivot as the datum)

or equivalently $V = mgL(1 - \cos\theta)$ with datum at the lowest point.

The Lagrangian is $L = T - V = \frac{1}{2}mL^2\theta^2 + mgL\cos\theta$.

(c) Define a holonomic constraint and state how the number of degrees of freedom (DOF) is determined for a system of N particles with k holonomic constraints. **[0.5]**

Ans:

A holonomic constraint is one that can be expressed as an equation relating only coordinates (and possibly time), not velocities: $f(q_1, q_2, \ldots, t) = 0$.

DOF = 3N − k, where N = number of particles, k = number of independent holonomic constraints.

Example: A bead on a wire (k = 2 of 3 coordinates constrained → 1 DOF).

(d) A spring-mass system has m = 4 kg and k = 256 N/m. Calculate the natural frequency ω_n (rad/s) and the time period T of free oscillation. **[0.5]**

Ans:

$\omega_n = \sqrt{(k/m)} = \sqrt{(256/4)} = \sqrt{64} = 8$ rad/s.

$T = 2\pi/\omega_n = 2\pi/8 = \pi/4 \approx 0.785$ s.

Solve any TWO from Q2, Q3, Q4 [3 Marks each = 6 Marks]

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO1, CO2

A simple pendulum consists of a point mass m = 1 kg attached to a massless, inextensible string of length L = 1 m. The bob swings in a vertical plane under gravity ($g = 9.81$ m/s²). Using the angle θ from the vertical as the generalized coordinate, analyze the system using the Lagrangian method. Use the following relations: Lagrangian $L = T - V$; Euler-Lagrange equation: $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$; for small angles: $\sin\theta \approx \theta$.

(a) State the number of degrees of freedom of this system and identify the generalized coordinate. **[0.5]**

Ans:

The pendulum has 1 degree of freedom (DOF = 1).

Generalized coordinate: θ — the angle of the string measured from the vertical.

(b) Write the position of the mass in Cartesian coordinates (x, y) in terms of θ and L. **[0.5]**

Ans:

Taking the pivot as the origin, with x horizontal and y vertically downward:

$x = L\sin\theta = 1.0\sin\theta$

$y = L\cos\theta = 1.0\cos\theta$

Differentiating: $\dot{x} = L\dot{\theta}\cos\theta$, $\dot{y} = -L\dot{\theta}\sin\theta$

(c) Derive the kinetic energy T in terms of θ, $\dot{\theta}$, m, and L. **[0.5]**

Ans:

$v^2 = \dot{x}^2 + \dot{y}^2 = L^2\theta^2\cos^2\theta + L^2\theta^2\sin^2\theta = L^2\theta^2$

$T = \frac{1}{2}mv^2 = \frac{1}{2} \times 1.0 \times 1.0^2 \times \theta^2 = \frac{1}{2}mL^2\theta^2$

$T = \frac{1}{2}(1.0)(1.0^2)\theta^2 = 0.5\,\theta^2$ J

(d) Write the potential energy V in terms of θ, m, g, and L. **[0.5]**

Ans:

Taking the pivot as the datum (y positive downward):

$V = -mgy = -mgL\cos\theta = -1.0 \times 9.81 \times 1.0 \times \cos\theta$

$V = -9.81\cos\theta$ J

(e) Form the Lagrangian L = T − V and apply the Euler-Lagrange equation to derive the equation of motion. **[0.5]**

Ans:

$L = T - V = \frac{1}{2}mL^2\theta^2 + mgL\cos\theta$

Euler-Lagrange: $d/dt(\partial L/\partial\dot{\theta}) - \partial L/\partial\theta = 0$

$\partial L/\partial\dot{\theta} = mL^2\dot{\theta} = 1.0 \times 1.0^2 \times \dot{\theta} = 1.0\,\dot{\theta}$

$d/dt(mL^2\dot{\theta}) = mL^2\ddot{\theta} = 1.0\,\ddot{\theta}$

$\partial L/\partial\theta = -mgL\sin\theta = -9.81\sin\theta$

EOM: $1.0\,\ddot{\theta} + 9.81\sin\theta = 0$

Simplifying (divide by mL^2): $\ddot{\theta} + (g/L)\sin\theta = 0$

$\rightarrow \ddot{\theta} + (9.81/1.0)\sin\theta = 0 \rightarrow \ddot{\theta} + 9.81\sin\theta = 0$

(f) For small oscillations ($\sin\theta \approx \theta$), determine the natural frequency of the pendulum. **[0.5]**

Ans:

For small angles, $\sin\theta \approx \theta \rightarrow \ddot{\theta} + (g/L)\theta = 0$

This is SHM with $\omega_n^2 = g/L = 9.81/1.0 = 9.81$

$\omega_n = \sqrt{(9.81)} = 3.132$ rad/s

Time period $T = 2\pi/\omega_n = 2\pi/3.132 = 2.006$ s

Q3 [3 Marks]

Bloom: BL 3–4 (Apply / Analyze) | CO2

A horizontal platform rotates at a constant angular velocity $\Omega = 4$ rad/s about a vertical axis. An object at a radial distance $r = 1.5$ m from the center moves radially outward at a constant speed $v_{rel} = 2$ m/s ($\dot{r} = 2$ m/s, $\ddot{r} = 0$). The angular acceleration of the platform is zero ($\alpha = 0$).

(a) Write the general expression relating the acceleration observed from the fixed (inertial) frame to quantities measured in the rotating frame. Identify each term. **[0.5]**

Ans:

The fixed-frame (inertial) acceleration, expressed in terms of rotating-frame quantities, is:

$a\square_{fixed} = a\square_{rel} + 2\Omega\square \times v\square_{rel} + \alpha\square \times r\square + \Omega\square \times (\Omega\square \times r\square)$

where:

$a\square_{rel}$ = acceleration of the object in the rotating frame ($\ddot{r} = 0$ here),

$2\Omega\square \times v\square_{rel}$ = Coriolis acceleration,

$\alpha\square \times r\square$ = Euler (angular acceleration) term (zero here since $\alpha = 0$),

$\Omega\square \times (\Omega\square \times r\square)$ = centripetal acceleration (directed radially inward).

(b) Calculate the centripetal acceleration magnitude at $r = 1.5$ m. **[0.5]**

Ans:

$a_{centripetal} = \Omega^2 r = 4.0^2 \times 1.5 = 16 \times 1.5 = 24.0$ m/s²

Direction: radially inward (toward the axis of rotation).

(c) Calculate the Coriolis acceleration magnitude. **[0.5]**

Ans:

$a_{Coriolis} = 2\Omega v_{rel} = 2 \times 4.0 \times 2.0 = 16.0$ m/s²

(d) State the direction of the Coriolis acceleration relative to the radial velocity. **[0.5]**

Ans:

By the right-hand rule for $2\Omega\square \times v\square_{rel}$ (Ω along vertical, v_{rel} radially outward): the Coriolis acceleration is perpendicular to v_{rel}, directed tangentially in the direction opposite to the platform's rotation.

(e) Calculate the total acceleration magnitude of the object as observed from the fixed (inertial) frame. **[0.5]**

Ans:

$|a_{total}| = \sqrt{a_{cent}^2 + a_{Cor}^2} = \sqrt{24.0^2 + 16.0^2}$

$= \sqrt{576 + 256} = \sqrt{832} = 28.84$ m/s²

(f) If the object continues at the same radial speed, find the centripetal acceleration when $r = 3$ m and comment on how it changes. **[0.5]**

Ans:

At $r = 3$ m: $a_{centripetal} = \Omega^2 r = 4.0^2 \times 3 = 48.0$ m/s²

Compared to 24.0 m/s² at $r = 1.5$ m, the centripetal acceleration has doubled. It increases linearly with radial distance (a_{cent} □ r), while the Coriolis acceleration (16.0 m/s²) remains unchanged since v_{rel} is constant.

Q4 [3 Marks]

Bloom: BL 4–5 (Analyze / Evaluate) | CO2, CO3

A mass $m = 3$ kg is attached to a spring of stiffness $k = 300$ N/m and a viscous damper with damping coefficient $c = 12$ N·s/m. The system undergoes free vibration after being displaced 0.05 m from equilibrium and released from rest. Take the equilibrium position as the origin.

(a) Write the equation of motion for this damped spring-mass system. **[0.5]**

Ans:

The equation of motion for a damped SDOF system:

$m\ddot{x} + c\dot{x} + kx = 0$

$3.0\ddot{x} + 12.0\dot{x} + 300.0x = 0$

Dividing by m: $\ddot{x} + 4.0\dot{x} + 100.0x = 0$

Or: $\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = 0$

(b) Calculate the natural frequency ω_n and the critical damping coefficient c_{cr}. **[0.5]**

Ans:

$\omega_n = \sqrt{k/m} = \sqrt{300.0/3.0} = \sqrt{100} = 10.00$ rad/s

$c_{cr} = 2\sqrt{km} = 2\sqrt{300.0 \times 3.0} = 2\sqrt{900} = 2 \times 30.00 = 60.00$ N·s/m

(c) Calculate the damping ratio ζ and state the type of damping. **[0.5]**

Ans:

$\zeta = c / c_{cr} = 12.0 / 60.00 = 0.2000$

Since $\zeta = 0.2000 < 1 \rightarrow$ the system is UNDERDAMPED.
It will oscillate with exponentially decaying amplitude.

(d) Calculate the damped natural frequency ω_d. **[0.5]**

Ans:

$\omega_d = \omega_n \sqrt{(1 - \zeta^2)} = 10.00 \times \sqrt{(1 - 0.2000^2)}$

$= 10.00 \times \sqrt{(1 - 0.040000)}$

$= 10.00 \times 0.979796$

$= 9.7980$ rad/s ≈ 9.80 rad/s

(e) Calculate the logarithmic decrement δ and find the ratio of two successive peak amplitudes. **[0.5]**

Ans:

$\delta = 2\pi\zeta / \sqrt{(1 - \zeta^2)} = 2\pi \times 0.2000 / \sqrt{(1 - 0.040000)}$

$= 1.2566 / 0.979796$

$= 1.2825$

Ratio of successive peaks: $x_n / x_{(n+1)} = e^{\delta} = e^{1.2825} = 3.6058$

Or equivalently $x_{(n+1)} / x_n = e^{(-\delta)} = 0.2773$

Each successive peak is $\approx$ 27.73% of the previous one.

(f) If the damper is removed ($c = 0$), write the displacement as a function of time $x(t)$ for the given initial conditions. **[0.5]**

Ans:

With $c = 0$ (undamped), EOM: $m\ddot{x} + kx = 0$

$x(t) = x_0 \cos(\omega_n t) = 0.05 \cos(10.00\, t)$ m

(Since initial conditions are $x(0) = 0.05$ m, $\dot{x}(0) = 0$, the solution is a pure cosine with amplitude 0.05 m and period $T = 2\pi/10.00 = 0.6283$ s)

— End of Set 1 Model Answers —

SVKM'S NMIMS — MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Re-Examination | Dynamic Systems Modeling | Set 2

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1

Answer the following:

(a) Write the expressions for the radial and transverse components of acceleration in polar coordinates (a_r and a_θ) and identify the centripetal term. **[0.5]**

Ans:

In polar coordinates the acceleration components are:

$a_r = \ddot{r} - r\dot{\theta}^2$ (radial component)

$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$ (transverse component)

The term $-r\dot{\theta}^2$ in a_r is the centripetal acceleration, directed radially inward. The term $2\dot{r}\dot{\theta}$ in a_θ is the Coriolis component arising from simultaneous radial and angular motion.

(b) State Hamilton's principle (principle of least action) and explain its relationship to the Euler-Lagrange equation. **[0.5]**

Ans:

Hamilton's principle states that the actual path taken by a mechanical system between two time instants t_1 and t_2 is the one that makes the action integral stationary:

$\delta S = \delta \int(t_1 \text{ to } t_2)\, L\, dt = 0$, where $L = T - V$.

Applying the calculus of variations to this condition yields the Euler-Lagrange equation:

$d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$.

(c) Differentiate between holonomic and non-holonomic constraints. Give one example of each from a mechanical system. **[0.5]**

Ans:

Holonomic constraint: can be expressed as $f(q_1, q_2, \ldots, t) = 0$ (depends only on coordinates and time, not velocities).

Example: A bead on a wire — its position is restricted to the wire curve.

Non-holonomic constraint: involves velocities or inequalities and cannot be reduced to a coordinate-only equation.

Example: A wheel rolling without slipping — the constraint $dx = R\, d\theta$ relates velocity components but cannot be integrated into a position-only equation for all paths.

(d) State the parallel axis theorem for moment of inertia and give its mathematical expression. **[0.5]**

Ans:

The parallel axis theorem states that the moment of inertia about any axis parallel to one through the center of mass is:

$I = I_{cm} + md^2$

where I_{cm} = moment of inertia about the CM axis, m = total mass, d = perpendicular distance between the two axes. I is always smallest about the CM axis.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

(a) A particle moves along a spiral with $r = 3$ m, $\dot{r} = 0$, and $\dot{\theta} = 2$ rad/s (constant). Calculate the centripetal acceleration and state its direction. **[0.5]**

Ans:

For circular motion with $r = 3$ m, $\dot{r} = 0$, $\dot{\theta} = 2$ rad/s:

$a_{centripetal} = r\dot{\theta}^2 = 3 \times 2^2 = 3 \times 4 = 12$ m/s^2

(In polar form: $a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 12 = -12$ m/s^2)

Direction: radially inward, toward the center of curvature.

(b) Explain the concept of generalized coordinates. For a double pendulum in a plane, state the number of DOF and identify the generalized coordinates. **[0.5]**

Ans:

Generalized coordinates are the minimum set of independent coordinates needed to completely describe a system's configuration. They need not be Cartesian — they can be angles, arc lengths, etc.

A double pendulum in a plane has 2 DOF.

Generalized coordinates: θ_1 (angle of first link) and θ_2 (angle of second link), both measured from the vertical.

(c) Define the Coriolis acceleration. State the expression for its magnitude and explain in which physical situations it becomes significant. **[0.5]**

Ans:

The Coriolis acceleration arises when an object moves with velocity v_rel in a reference frame rotating at angular velocity Ω:

$a_{Coriolis} = 2\Omega \times v_{rel}$, magnitude $= 2\Omega v_{rel}$

It is perpendicular to both Ω and v_rel. It becomes significant in large-scale meteorology (weather systems), long-range ballistics, and fast-rotating machinery where the product Ωv_{rel} is comparable to other acceleration terms.

(d) A uniform rod of mass m = 3 kg and length L = 1.2 m is pivoted at one end. Calculate its moment of inertia about the pivot using the parallel axis theorem. **[0.5]**

Ans:

For a uniform rod: $I_{cm} = (1/12)mL^2 = (1/12) \times 3 \times 1.2^2 = (1/12) \times 3 \times 1.44 = 0.36$ kg·m².

Parallel axis theorem: $I_{pivot} = I_{cm} + m(L/2)^2$

$= 0.36 + 3 \times (0.6)^2 = 0.36 + 3 \times 0.36 = 0.36 + 1.08$

$= 1.44$ kg·m².

(Check: $I_{pivot} = (1/3)mL^2 = (1/3) \times 3 \times 1.44 = 1.44$ kg·m² □)

Solve any TWO from Q2, Q3, Q4 [3 Marks each = 6 Marks]

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO1, CO2

A simple pendulum consists of a point mass m = 2 kg attached to a massless, inextensible string of length L = 0.5 m. The bob swings in a vertical plane under gravity ($g = 9.81$ m/s²). Using the angle θ from the vertical as the generalized coordinate, analyze the system using the Lagrangian method. Use the following relations: Lagrangian $L = T - V$; Euler-Lagrange equation: $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$; for small angles: $\sin\theta \approx \theta$.

(a) State the number of degrees of freedom of this system and identify the generalized coordinate. **[0.5]**

Ans:

The pendulum has 1 degree of freedom (DOF = 1).

Generalized coordinate: θ — the angle of the string measured from the vertical.

(b) Write the position of the mass in Cartesian coordinates (x, y) in terms of θ and L. **[0.5]**

Ans:

Taking the pivot as the origin, with x horizontal and y vertically downward:

$x = L \sin\theta = 0.5 \sin\theta$

$y = L \cos\theta = 0.5 \cos\theta$

Differentiating: $\dot{x} = L\dot{\theta} \cos\theta$, $\dot{y} = -L\dot{\theta} \sin\theta$

(c) Derive the kinetic energy T in terms of θ, $\dot{\theta}$, m, and L. **[0.5]**

Ans:

$v^2 = \dot{x}^2 + \dot{y}^2 = L^2\dot{\theta}^2\cos^2\theta + L^2\dot{\theta}^2\sin^2\theta = L^2\dot{\theta}^2$

$T = ½mv^2 = ½ \times 2.0 \times 0.5^2 \times \dot{\theta}^2 = ½mL^2\dot{\theta}^2$

$T = ½(2.0)(0.5^2)\dot{\theta}^2 = 0.25\ \dot{\theta}^2$ J

(d) Write the potential energy V in terms of θ, m, g, and L. **[0.5]**

Ans:

Taking the pivot as the datum (y positive downward):

$V = -mgy = -mgL \cos\theta = -2.0 \times 9.81 \times 0.5 \times \cos\theta$

$V = -9.81 \cos\theta$ J

(e) Form the Lagrangian $L = T - V$ and apply the Euler-Lagrange equation to derive the equation of motion. **[0.5]**

Ans:

$L = T - V = ½mL^2\dot{\theta}^2 + mgL \cos\theta$

Euler-Lagrange: $d/dt(\partial L/\partial\dot{\theta}) - \partial L/\partial\theta = 0$

$\partial L/\partial\dot{\theta} = mL^2\dot{\theta} = 2.0 \times 0.5^2 \times \dot{\theta} = 0.5\ \dot{\theta}$

$d/dt(mL^2\dot{\theta}) = mL^2\ddot{\theta} = 0.5\ \ddot{\theta}$

$\partial L/\partial\theta = -mgL \sin\theta = -9.81 \sin\theta$

EOM: $0.5\ \ddot{\theta} + 9.81 \sin\theta = 0$

Simplifying (divide by mL^2): $\ddot{\theta} + (g/L) \sin\theta = 0$

$\rightarrow \ddot{\theta} + (9.81/0.5) \sin\theta = 0 \rightarrow \ddot{\theta} + 19.62 \sin\theta = 0$

(f) For small oscillations ($\sin\theta \approx \theta$), determine the natural frequency of the pendulum. **[0.5]**

Ans:

For small angles, $\sin\theta \approx \theta \rightarrow \ddot{\theta} + (g/L)\theta = 0$

This is SHM with $\omega_n^2 = g/L = 9.81/0.5 = 19.62$

$\omega_n = \sqrt{19.62} = 4.429$ rad/s

Time period $T = 2\pi/\omega_n = 2\pi/4.429 = 1.419$ s

Q3 [3 Marks]

Bloom: BL 3–4 (Apply / Analyze) | CO2

A horizontal turntable rotates at a constant angular velocity $\Omega = 5$ rad/s about a vertical axis. A small ball at $r = 2$ m from the center is pushed radially outward at a constant relative speed $v_{rel} = 3$ m/s ($\dot{r} = 3$ m/s, $\ddot{r} = 0$). The angular acceleration of the turntable is zero ($\alpha = 0$).

(a) Write the general expression relating the acceleration observed from the fixed (inertial) frame to quantities measured in the rotating frame. Identify each term. **[0.5]**

Ans:

The fixed-frame (inertial) acceleration, expressed in terms of rotating-frame quantities, is:

$\vec{a}_{fixed} = \vec{a}_{rel} + 2\vec{\Omega} \times \vec{v}_{rel} + \vec{\alpha} \times \vec{r} + \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$

where:

$\vec{a}_{rel}$ = acceleration of the object in the rotating frame ($\ddot{r} = 0$ here),

$2\vec{\Omega} \times \vec{v}_{rel}$ = Coriolis acceleration,

$\vec{\alpha} \times \vec{r}$ = Euler (angular acceleration) term (zero here since $\alpha = 0$),

$\vec{\Omega} \times (\vec{\Omega} \times \vec{r})$ = centripetal acceleration (directed radially inward).

(b) Calculate the centripetal acceleration magnitude at $r = 2$ m. **[0.5]**

Ans:

$a_{centripetal} = \Omega^2 r = 5.0^2 \times 2.0 = 25 \times 2.0 = 50.0$ m/s²

Direction: radially inward (toward the axis of rotation).

(c) Calculate the Coriolis acceleration magnitude. **[0.5]**

Ans:

$a_{Coriolis} = 2\Omega v_{rel} = 2 \times 5.0 \times 3.0 = 30.0$ m/s²

(d) State the direction of the Coriolis acceleration relative to the radial velocity. **[0.5]**

Ans:

By the right-hand rule for $2\vec{\Omega} \times \vec{v}_{rel}$ (Ω along vertical, v_{rel} radially outward): the Coriolis acceleration is perpendicular to v_{rel}, directed tangentially in the direction opposite to the platform's rotation.

(e) Calculate the total acceleration magnitude of the object as observed from the fixed (inertial) frame. **[0.5]**

Ans:

$|a_{total}| = \sqrt{a_{cent}^2 + a_{Cor}^2} = \sqrt{50.0^2 + 30.0^2}$

$= \sqrt{2500 + 900} = \sqrt{3400} = 58.31$ m/s²

(f) If the ball continues at the same radial speed, find the centripetal acceleration when $r = 4$ m and comment on how it changes. **[0.5]**

Ans:

At $r = 4$ m: $a_{centripetal} = \Omega^2 r = 5.0^2 \times 4 = 100.0$ m/s²

Compared to 50.0 m/s² at $r = 2.0$ m, the centripetal acceleration has doubled. It increases linearly with radial distance ($a_{cent} \propto r$), while the Coriolis acceleration (30.0 m/s²) remains unchanged since v_{rel} is constant.

Q4 [3 Marks]

Bloom: BL 4–5 (Analyze / Evaluate) | CO2, CO3

A uniform rod of mass $m = 2$ kg and length $L = 1.2$ m is pivoted at one end and oscillates as a physical (compound) pendulum under gravity. Take $g = 9.81$ m/s². Use: $I_{rod\_cm} = (1/12)mL^2$; parallel axis theorem: $I = I_{cm} + md^2$; period of physical pendulum: $T = 2\pi\sqrt{I_{pivot} / (mgd)}$, where d = distance from pivot to center of mass.

(a) Calculate the moment of inertia of the rod about its center of mass (I_{cm}). **[0.5]**

Ans:

$I_{cm} = (1/12)mL^2 = (1/12) \times 2.0 \times 1.2^2$

$= (1/12) \times 2.0 \times 1.44 = 0.2400$ kg·m²

(b) Using the parallel axis theorem, calculate the moment of inertia about the pivot at one end (I_{pivot}). **[0.5]**

Ans:

$I_{pivot} = I_{cm} + m(L/2)^2 = 0.2400 + 2.0 \times (1.2/2)^2$

$= 0.2400 + 2.0 \times 0.6^2 = 0.2400 + 0.7200$

$= 0.9600$ kg·m²

(Check: $I_{pivot} = (1/3)mL^2 = (1/3) \times 2.0 \times 1.44 = 0.9600$ kg·m² □)

(c) Find the distance d of the center of mass from the pivot. **[0.5]**

Ans:

$d = L/2 = 1.2/2 = 0.6$ m

(The center of mass of a uniform rod is at its midpoint.)

(d) Write the equation of motion for small-angle oscillations and determine the natural frequency ω_n. **[0.5]**

Ans:

Restoring torque: $\tau = -mgd \sin\theta \approx -mgd\theta$ (small angle)

Equation of motion: $I_{pivot}\ddot{\theta} + mgd\,\theta = 0$

$\rightarrow \ddot{\theta} + (mgd / I_{pivot})\,\theta = 0$

$\omega_n = \sqrt{(mgd / I_{pivot})} = \sqrt{(2.0 \times 9.81 \times 0.6 / 0.9600)}$

$= \sqrt{(11.7720 / 0.9600)} = \sqrt{12.2625}$

$= 3.5018$ rad/s ≈ 3.502 rad/s

(e) Calculate the time period of oscillation T. **[0.5]**

Ans:

$T = 2\pi / \omega_n = 2\pi / 3.5018 = 1.7943$ s ≈ 1.794 s

(f) Compare this period with that of a simple pendulum of the same length ($L = 1.2$ m) and explain the difference. **[0.5]**

Ans:

Simple pendulum of same length $L = 1.2$ m:

$T_{simple} = 2\pi\sqrt{(L/g)} = 2\pi\sqrt{(1.2/9.81)} = 2\pi \times 0.3497$

$= 2.1975$ s ≈ 2.198 s

Physical pendulum period (1.794 s) < Simple pendulum period (2.198 s).

This is because the physical pendulum's mass is distributed along its length, with some mass closer to the pivot than L. The effective pendulum length $L_{eff} = I_{pivot}/(md) = 0.9600/(2.0\times0.6) = 0.8000$ m < 1.2 m.

— End of Set 2 Model Answers —